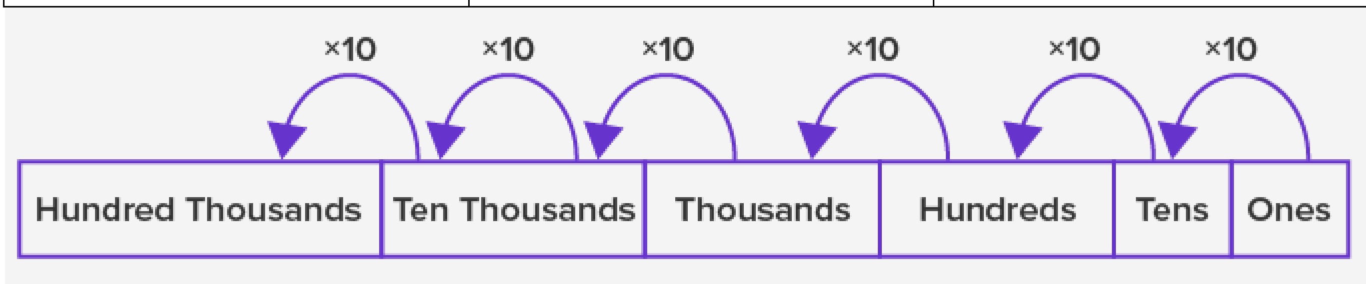
Number Systems

Decimal Number System:
The decimal number system consists of 10 digits 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 and is the most commonly used number system. We use the combination of these 10 digits to form all other numbers. The value of a digit in a number depends upon its position in the number.

Each place to the left is ten times greater than the place to its right, that is, as we move from the right to the left, the place value increases ten times with each place.

Decimal	Binary	Hexadecimal
0	0	0
1	1	1
2	10	2
3	11	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000	8
9	1001	9
10	1010	10
11	1011	A
12	1100	B
13	1101	C
14	1110	D
15	1111	E
16	10000	F
17	10001	11
18	10010	12
19	10011	13
20	10100	14



- A decimal number system is also called the Base 10 system.
- A number 49,365 is read as Forty-nine thousand three hundred sixty-five, where the value of 4 is forty thousand, 9 is nine thousand, 3 is three hundred, 6 is sixty and 5 is five.

Binary Number System

In the binary number system, we only use two digits 0 and 1. It means a 2 number system.
Example of binary numbers: 1011; 101010; 1101101. Each digit in a binary number is called a bit. So, a binary number 101 has 3 bits. 499787080

Computers and other digital devices use the binary system. The binary number system uses Base 2.

Hexadecimal Number System

The word hexadecimal comes from Hexa meaning 6, and decimal meaning 10. So, in a hexadecimal number system, there are 16 digits. It consists of digits 0 to 9 and then has first 5 letters of the alphabet as:

Decimal	0	1	2	3	4	5	6	7	8	9					
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E
Binary	0	1													

The table below shows numbers 1 to 20 using decimal, binary and hexadecimal numbers.
Examples:
Example of Decimal Number System:
The decimal number 1457 consists of the digit 7 in the units position, 5 in the tens place, 4 in the hundreds position, and 1 in the thousands place whose value can be written as:

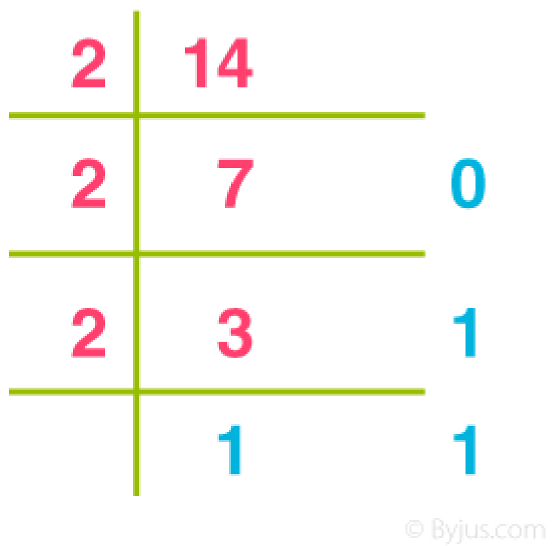
$(1\times10^3) + (4\times10^2) + (5\times10^1) + (7\times10^0)$

$(1\times1000) + (4\times100) + (5\times10) + (7\times1)$

$1000 + 400 + 50 + 7$

1457

Example of Binary Number System:
Write $(14)_{10}$ as a binary number.
Solution:



Base 2 Number System Example

$\therefore (14)_{10} = 1110_2$

Octal Number System (Base 8 Number System)

In the octal number system, the base is 8 and it uses numbers from 0 to 7 to represent numbers. Octal numbers are commonly used in computer applications. Converting an octal number to decimal is the same as decimal conversion and is explained below using an example.

Example: Convert 215_8 into decimal.

Solution:

$215_8 = 2 \times 8^2 + 1 \times 8^1 + 5 \times 8^0$

$= 2 \times 64 + 1 \times 8 + 5 \times 1$

$= 128 + 8 + 5$

$= 141_{10}$